A formal renaissance of the Blockchain trilemma

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1 Introduction

This is offered as a rebuttal to the argument posed by Craig S. Wright[1].

> The core structural failure of the so-called blockchain trilemma is its attempt to formulate a triadic trade-off across properties that belong to disjoint ontological categories. That is, the trilemma equivocates between properties that are semantically incommensurable and formally non interacting. This constitutes a category error in the strictest sense.

In phrase, an invariance preservation, a computational application process, and a vertex-edge property of a structural graph can coexist, as they are describing different things.

Are they?

If we assume the invariance describes the ongoing application to the structural graph, clearly not. As,

$$\mathcal{P} := check_{geq1}((\lambda, \kappa)G) : Time_{n+1} \ [\delta(q, \sqrt[n]{\sigma}/\Sigma)]Time_n$$

Naturally, the cyrptographic representation secured by $\mathcal{P}$ is demonstrated by n.

Simply allowing the hub to be specified with the decentralized properties $(\lambda, \kappa)G$ gives us a check light that holds to true. If we ask $\mathcal{P}$ we return a held property with an associated progressive application.

Within this broad setwise and associative structure we note something useful: if we associate *check* with the header-foot, $Time^{(-1)}$ with the verification and q with the message we may recover a ledger.

It expects that if $\mathcal{P}$ is held true as a given security protocol that we may not always satisfy, and that if other sources that the blockchain trilemma is more than semantics, are true, then the blockchain trilemma remains.

[1] https://arxiv.org/pdf/2507.05809v1